COMPLIMENTS

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REFERENCE BOOKS

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- Digital Design, M. Morris Mano, Michael D. Ciletti, Pearson Education, Inc.
- Digital Circuits and Design, S. Salivahanan, S. Arivazhagan, Oxford University Press.
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COMPLIMENTS

COMPLEMENTS:

- Complements are used in digital computers to simplify the subtraction operation and for logical manipulation.
- There are two types of complements for each base-r system.
- (i) the radix complement and is referred as r's complement. (ii) diminished radix complement and is referred as (r-1)'s complement.
- For binary numbers two types of complements are 2's complement and 1's complement.
- For decimal numbers two types of complements are 10's complement and 9's complement.

Reference: Digital Design, M. Morris Mano, Michael D. Ciletti

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DIMINISHED RADIX COMPLEMENT: Given positive number ‘N’ in base ‘r’ having an integer part of ‘n’ digits and fraction part of ‘m’ digits, then (r-1)’s complement of N is defined as

$$(r^n - r^{-m} - N)$$

For only integer part m = 0, then

(r-1)’s complement of N = $[(r^n-1)-N]$

- For decimal system, base (r =10), then r – 1 = 10 – 1 = 9
- Therefore, for decimal system (10-1)’s complement is called 9’s complement.
- For binary system r = 2, (r-1) = (2-1)’s complement is called 1’s complement.
- 9’s complement of a decimal system is obtained by subtracting each digit from 9.

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For example, 9's complement of 546700 = 999999 – 546700 = 453299

Similarly, 1's complement of a binary system is obtained by changing 1 to 0 and 0 to 1.

For example, 1's complement of 1011000 = 0100111.

Similarly, (r- 1)'s complement of octal system is called 7's complement and subtract each digit from 7.

For hexadecimal number (16 -1)'s complement is called 15's complement and subtract each digit from 15 (F).

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RADIX COMPLEMENT:

Given positive number 'N' in base 'r' having an integer part of 'n' digits then r's complement of N is defined as

$$(r^n - N) \text{ for } N \neq 0 \text{ and } 0 \text{ for } N = 0$$

- For decimal number system r = 10, then r's complement is called 10's complement.
- For binary system r = 2, then it is called 2's complement.
- For octal system r = 8, then it is called 8's complement.
- For hexadecimal system r = 16, then it is called 16's complement.

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10’s complement can be formed by subtracting the first non-zero least significant digit from 10, and subtracting all higher significant digits from 9.

For example, 10’s complement of 012398 = 987602

Similarly, 2’s complement can be formed by leaving all least significant 0 and the first 1 unchanged and replacing 1 with 0 and 0 with 1 in all other higher significant digits. For example,

2’s complement of $1101\underbrace{100} = 0010100$

Unchanged

2’s complement of 1111111 = 0000001

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NOTE: The 2's complement is the binary number that results when we add 1 to the 1's complement.

2's complement = 1's complement + 1

2's complement form is used to represent negative numbers.

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EXERCISE:

Number	1's complement	2's complement
10000000		

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EXERCISE:

Number	1's complement	2's complement
10000000	01111111	10000000

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EXERCISE:

Number	1's complement	2's complement
11011010		

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EXERCISE:

Number	1's complement	2's complement
11011010	00100101	00100110

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EXERCISE:

Number	1's complement	2's complement
10000101		

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EXERCISE:

Number	1's complement	2's complement
10000101	01111010	01111011

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EXERCISE:

Number	1's complement	2's complement
01110110		

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EXERCISE:

Number	1's complement	2's complement
01110110	10001001	10001010

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EXERCISE:

Number	1's complement	2's complement
10101010		

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EXERCISE:

Number	1's complement	2's complement
10101010	01010101	01010110

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EXERCISE:

Number	1's complement	2's complement
10000000	01111111	10000000
11011010	00100101	00100110
10000101	01111010	01111011
01110110	10001001	10001010
10101010	01010101	01010110

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EXERCISE:

Number	9's complement	10's complement
12345678		

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Number	9's complement	10's complement
12345678	87654321	87654322

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Number	9’s complement	10’s complement
24681234		

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Number	9's complement	10's complement
24681234	75318765	75318766

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Number	9's complement	10's complement
52784630		

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Number	9’s complement	10’s complement
52784630	47215369	47215370

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Number	9's complement	10's complement
25000000		

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Number	9's complement	10's complement
25000000	74999999	75000000

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Number	9’s complement	10’s complement
63325600		

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Number	9's complement	10's complement
63325600	36674399	36674400

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Number	9’s complement	10’s complement
12345678	87654321	87654322
24681234	75318765	75318766
52784630	47215369	47215370
25000000	74999999	75000000
63325600	36674399	36674400

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SUBTRACTION WITH COMPLEMENTS:

(A) 1’S COMPLEMENT SUBTRACTION METHOD: Here, instead of subtracting a number, we add 1’s complement of the number. There are two cases

(i) CASE 1: subtraction of smaller number from larger number:

Step 1: Determine 1’s complement of the smaller number (subtrahend)

Step 2: Add 1’s complement to the larger number (minuend) and the result is a positive number.

Step 3: Carry will be generated after the addition which is called End-Around Carry (EAC). Remove EAC and add it to the result.

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EXAMPLE: Subtract 101011 *(43)* from 111101 *(61)* using 1's complement.

Here, M = 111101 and S = 101011

1's complement of S = 010100

M	1 1 1 1 0 1		61
S	+ 0 1 0 1 0 0		- 43
	$\boxed{1}$ 01 0 0 0 1		18
EAC	1		
	0 1 0 0 1 0		

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CASE 2: subtraction of larger number from smaller number:

Step 1: Determine 1's complement of larger number (subtrahend)

Step 2: Add 1's complement to the smaller number (minuend).

Step 3: After addition, no carry will be generated but the answer is in 1's complement (negative number). To get the answer in true form, take the 1's complement of it and assign negative sign to the answer.

EXAMPLE: Subtract 111101 (S) from101011 (M) using 1's complement.

M (43)	1	0	1	0	1	1
1's complement of S (61)	0	0	0	0	1	0
Sum	1	0	1	1	0	1

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There is no carry. Therefore,

M – S = - (1’s complement of sum)

= - (1’s complement of 101101) = - 010010 (-18)

Advantages of 1’s complement subtraction:

1. The 1’s complement subtraction can be accomplished with a binary adder. Therefore, this method is useful in arithmetic logic unit.
2. It is very easy to find 1’s complement of a number.

Disadvantages:

1. In 1’s complement method, hardware implementation is difficult and it gives the concepts of negative zero.
2. It also gives one value less to represent negative values.

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2'S COMPLEMENT SUBTRACTION METHOD: Like 1's complement subtraction in 2's complement subtraction also, the subtraction is accomplished by addition.

CASE 1: Subtraction of Smaller Number from Larger Number:

Step 1: Determine the 2's complement of the smaller number.

Step 2: Add 2's complement to the larger number (minuend).

Step 3: Discard the carry generated.

EXAMPLE: If X = 1010100 (84), Y = 1000011 (67). Find X - Y using 2's complement.

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X		1	0	1	0	1	0	0
2's complement of Y		0	1	1	1	1	0	1
Sum	$\boxed{1}$	0	0	1	0	0	0	1
Discard End Carry	-1	0	0	0	0	0	0	0
X - Y		0	0	1	0	0	0	1

X – Y = 0 0 1 0 0 0 1

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CASE 2: Subtraction of Larger Number from Smaller Number:

Step 1: Determine the 2's complement of the larger number.

Step 2: Add 2's complement to the smaller number (minuend).

Step 3: Answer is in 2's complement form. To get answer in the true value form, take the 2's complement and assign negative sign to the answer.

Example: If X = 1010100 (84), Y = 1000011 (67), find Y – X using 2's complement.

Y	1	0	0	0	0	1	1
1's complement of X	0	1	0	1	1	0	0
Sum	1	1	0	1	1	1	1

There is no end carry. Therefore Y – X = - (2's complement of 1101111)
Y – X = - 0010001

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Example: Perform subtraction 1001 (9) - 101000 (40), using 2's complement.

Solution:

M (9)	0	0	1	0	0	1
2's complement of S (40)	0	1	1	0	0	0
Sum	1	0	0	0	0	1

There is no end carry, M – S = - (2's complement of 100001) = - (011111)

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SIGNED BINARY NUMBERS:

- If the binary is signed, then the leftmost bit represents the sign and the rest of the bits represent the number.
- If the binary number is assumed to be unsigned, then leftmost bit is significant bit of the number.
- Positive number always start with ‘0’ and the negative number starts from ‘1’ (leftmost).
- There are three ways in which signed numbers may be expressed:

i. Signed magnitude

ii. One’s complement

iii. Two’s complement

- In an 8 bit word, signed magnitude representation places the absolute value of the number in the 7 bits to the right of the sign bit.

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For example,

Sign bit

+7	0	0	0	0	0	1	1	1

Consider, other example

0	1	1	1	1	1	1	1	+127

Therefore, the biggest positive number for eight bit binary number is +127.

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For negative number take 2's complement of that number. Consider +7, take its 2's complement 1 1 1 1 1 0 0 1 → (-7)

signed bit

+7		0	0	0	0	0	1	1	1
-7		1	1	1	1	1	0	0	1
Sum	1	0	0	0	0	0	0	0	0
Discard End Carry	-1	0	0	0	0	0	0	0	0
		0	0	0	0	0	0	0	0

Therefore, for any positive number, its negative number will be the 2's complement of that positive number.

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NEGATION: Negation is the operation of converting a positive number to its negative equivalent or a negative number to its positive equivalent.

When signed binary numbers are represented in the 2's complement, negation is performed simply by performing the 2's complement operation.

For example,

	0	1	0	0	1	+ 9
2's complement (negate)	1	0	1	1	1	-9
Negate again	0	1	0	0	1	+ 9

So, we negate a signed binary number by 2's complementing it. This negation changes the number to its equivalent if opposite sign.

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EXAMPLE: Each of the following numbers is a signed binary number in the 2's complement system. Determine the decimal value in each case.

(i) 0 1 1 0 0 (ii) 1 1 0 1 0 (iii) 1 0 0 0 1

SOLUTION: (i) 0 1 1 0 0

The sign bit is '0', so the number is positive and four bits (1100) represent the true magnitude of the number.

$(1100)_2 = (12)_{10}$

Required number is + 12

(ii) 1 1 0 1 0 , here the sign bit is 1, so the number is negative but we cannot tell what is the magnitude. We can find the number as " find the

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magnitude by negating (2's complement) the number to convert it to its positive value.

2's complement of (1 1 0 1 0) = 0 0 1 1 0

$(0\ 0\ 1\ 1\ 0\ )_2 = +\ 6$

Since the result of negation is 0 0 1 1 0 = + 6, the original number will be – 6

(iii) 1 0 0 0 1

solution: The sign bit is 1, the number is negative.

2's complement of 1 0 0 0 1 = 0 1 1 1 1 (+15)

$\therefore$ Original number = 2's complement of (0 1 1 1 1) = 1 0 0 0 1 (- 15)

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SPECIAL CASE IN 2'S COMPLEMENT REPRESENTATION:

Whenever a signed number has 1 in the sign bit and all zeros for magnitude bits, its decimal equivalent is -2^N, where N is the number of bits in the magnitude. For example,

Sign bit *Magnitude bits*

$1\underbrace{000}$	$-2^3 = -8$
10000	$-2^4 = -16$
100000	$-2^5 = -32$ and so on

Thus, complete range of values in 2's complement system having N magnitude,

$$-2^N \text{ to } +(2^N - 1)$$

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EXAMPLE 1: What is the range of unsigned decimal values that can be represented by 8 bits?

Solution: with no sign bit, all 8 bits are used for the magnitude. Therefore, the values will be range from

$$(0\,0\,0\,0\,0\,0\,0\,0) = (0)_{10} \quad \text{to} \quad (1\,1\,1\,1\,1\,1\,1\,1)_2 = (255)_{10}$$

EXAMPLE 2: What is the range of signed decimal value that can be represented by 8 bits?

Solution: The largest negative value is

$$(1\,0\,0\,0\,0\,0\,0\,0)_2 = -2^7 = -128_{10}$$

The largest positive value is

$$(0\,1\,1\,1\,1\,1\,1\,1)_2 = + 2^7 - 1 =127_{10}$$

Thus range is from – 128 to + 127. this is toatl 256 different values, including zero.

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EXAMPLE 3: (a) What range of signed decimal values can be represented using 12 bits including sign bit?

(b) How many bits would be required to represent decimal numbers from – 32768 to + 32767?

Solution: (a) The largest negative value of 12 bits is

(1 0 0 0 0 0 0 0 0 0 0) = -211 = - 2048

The largest value of 12 bits is

(0 1 1 1 1 1 1 1 1 1 1 1) = + 211 – 1 = 2047

The range will be from – 2048 to + 2047

(b) The given range is from – 32768 to + 32767

Total = 32768 + 32767 = 65535 = 2^{15}

N = no. of magnitude bits = 15

Including sign bit = 15 + 1 = 16

THANK YOU

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